Direct and iterative methods for linear systems

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Linear system of equations

In many real-world applications, we need to solve linear system of n equations with n variables $x_1, \dots, x_n$:

$$\begin{aligned}
E_1 : \qquad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\
E_2 : \qquad & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\
& \qquad\qquad\qquad\qquad\qquad\quad \vdots \\
E_n : \qquad & a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n
\end{aligned}$$

We're given $a_{ij},\ 1 \leq i, j \leq n$ and $b_i,\ (1 \leq i \leq n)$, and want to find $x_1, \dots, x_n$ that satisfy the n equations $E_1, \dots, E_n$.

Linear system of equations

General approach: Gauss elimination.

We use three operations to simplify the linear system:

- Equation E_i can be multiplied by λE_i for any $\lambda \neq 0$: $\lambda E_i \to E_i$
- E_j is multiplied by λ and added to E_i: $\lambda E_j + E_i \to E_i$
- Switch E_i and E_j: $E_i \leftrightarrow E_j$

The goal is to simply the linear system into a triangular form, and apply backward substitution to get $x_1, \ldots, x_n$.

Linear system of equations

Generally, we form the augmented matrix

$$\tilde{A} = [A \ \mathbf{b}], \quad \text{where } A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \text{ and } \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

and apply Gaussian elimination to get a triangular form of $\tilde{A}$ then apply backward substitution. Total cost is $O(n^3)$.

Pivoting strategies

Standard Gauss elimination may not work properly in numerical computations.

Example

Apply Gauss elimination to the system

$$\begin{aligned} E_1: &\quad 0.003000x_1 + 59.14x_2 = 59.17 \\ E_2: &\quad 5.291x_1 - 6.130x_2 = 46.78 \end{aligned}$$

with $\underline{\textit{four digits}}$ *for arithmetic rounding. Compare the result to exact solution* $x_1 = 10.00$ *and* $x_2 = 1.000$*.*

Pivoting strategies

Solution: We need to multiply E_1 by $\frac{5.291}{0.003000} = 1763.66\bar{6} \approx 1764$, then subtract it from E_2 and get:

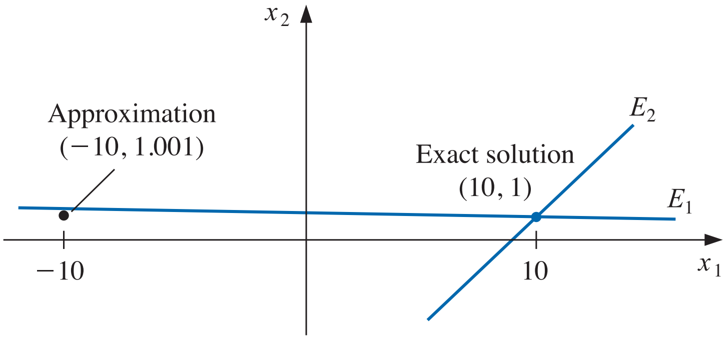
$$\begin{aligned} 0.003000x_1 + 59.14x_2 &\approx 59.17 \\ -104300x_2 &\approx -104400 \end{aligned}$$

On the other hand, the exact system without rounding error:

$$\begin{aligned} 0.003000x_1 + 59.14x_2 &\approx 59.17 \\ -104309.37\bar{6}x_2 &\approx -104309.37\bar{6} \end{aligned}$$

Solving the former yields $x_2 = 1.001$ (still close to exact solution 1.000), but $x_1 = \frac{59.17-59.14x_2}{0.003000} = -10.00$ (far from exact solution 10.00).

Pivoting strategies



Partial pivoting

- The issue above comes up because the pivot a_{kk} is smaller than the remaining a_{ij} $(i, j > k)$.
- One remedy, called **partial pivoting**, is interchanging rows k and p (where $|a_{ik}| = \max\{|a_{ik}| : i = k, \dots, n\}$).
- Sometimes interchange columns can also be performed.

For example, when we are about to do pivoting for the k-th time (i.e., $a_{kk}x_k$ term), we switch row p and current row k so that

$$p = \arg\max_{k \le i \le n} |a_{ik}|$$

Redo the example above, we will get exact solution.

Scaled partial pivoting

Consider the following example:

$$
\begin{aligned}
E_1: \quad & 30.00x_1 + 591400x_2 = 591700 \\
E_2: \quad & 5.291x_1 - 6.130x_2 = 46.78
\end{aligned}
$$

This is equivalent to example above, except that E_1 is multiplied by 10^4.

If we apply partial pivoting above, we will not exchange E_1 and E_2 since $30.00 > 5.291$, and will end up with the same *incorrect* answer $x_2 = 1.001$ and $x_1 = -10.00$.

To overcome this issue, we can scale the coefficients of each row i by $1/s_i$ where $s_i = \max_{1\le j\le n} |a_{ij}|$. Then apply partial pivoting based on the scaled values.

Scaled partial pivoting

Applying scaled partial pivoting to the example above, we first have

$s_1 = \max\{30.00, 519400\} = 519400, s_2 = \max\{5.291, 6.130\} = 6.130$

Hence we get $\frac{a_{11}}{s_1} = \frac{30.00}{519400} \approx 0.5073 \times 10^{-4}$, and $\frac{a_{21}}{s_2} = \frac{5.291}{6.130} = 0.8631$, the others are ± 1. By comparing $\frac{a_{11}}{s_1}$ and $\frac{a_{21}}{s_2}$, we will exchange E_1 and E_2, and hence obtain

$$E_1: \quad 5.291x_1 - 6.130x_2 = 46.78$$
$$E_2: \quad 30.00x_1 + 591400x_2 = 591700$$

and apply Gauss elimination to obtain correct answer $x_2 = 1.000$ and $x_1 = 10.00$.

Complete pivoting

For each of the n steps, find the largest magnitude among all coefficients a_{ij} for $k \leq i, j \leq n$. Then switch rows and/or columns so that the one with largest magnitude is in the pivot position.

This requires $O(n^3)$ comparisons. Only worth it if the accuracy improvement justifies the cost.

Linear algebra: quick review

We call A an $m \times n$ (m-by-n) **matrix** if A is an array of mn numbers with m rows and n columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

We may simply denote it by $A = [a_{ij}]$, when its size is clear in the context.

Linear algebra: quick review

- We call two matrices equal, i.e., $A = B$, if $a_{ij} = b_{ij}$ for all i, j.
- The sum of two matrices of same size is: $A \pm B = [a_{ij} \pm b_{ij}]$.
- Scalar multiplication of A by $\lambda \in \mathbb{R}$ is $\lambda A = [\lambda a_{ij}]$.
- We denote the matrix of all zeros by 0, and $-A = [-a_{ij}]$.

Linear algebra: quick review

The set of all $m \times n$ matrices forms a **vector space**:

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0 = 0 + A$
- $A + (-A) = 0$
- $\lambda(A + B) = \lambda A + \lambda B$
- $(\lambda + \mu)A = \lambda A + \mu A$
- $\lambda(\mu A) = (\lambda\mu)A$
- $1A = A$

Linear algebra: quick review

For matrix A of size $m \times n$ and (column) vector b of dimension n, we define the matrix-vector multiplication (product) by

$$Ab = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^n a_{1j} b_j \\ \sum_{j=1}^n a_{2j} b_j \\ \vdots \\ \sum_{j=1}^n a_{mj} b_j \end{bmatrix}$$

Linear algebra: quick review

For matrix A of size $m \times n$ and matrix B of size $n \times k$, we define the matrix-matrix multiplication (product) by

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nk} \end{bmatrix}$$
$$= \begin{bmatrix} \sum_{j=1}^n a_{1j}b_{j1} & \sum_{j=1}^n a_{1j}b_{j1} & \cdots & \sum_{j=1}^n a_{1j}b_{jk} \\ \sum_{j=1}^n a_{2j}b_{j1} & \sum_{j=1}^n a_{2j}b_{j1} & \cdots & \sum_{j=1}^n a_{2j}b_{jk} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n a_{mj}b_{j1} & \sum_{j=1}^n a_{mj}b_{j1} & \cdots & \sum_{j=1}^n a_{mj}b_{jk} \end{bmatrix} \in \mathbb{R}^{m\times k}$$

That is, if $C = AB$, then $[c_{ij}] = [\sum_r a_{ir}b_{rj}]$ for all i, j.

Linear algebra: quick review

Some properties of matrix product

- $A(BC) = (AB)C$
- $A(B + D) = AB + AD$
- $\lambda(AB) = (\lambda A)B = A(\lambda B)$

Remark

Note that $AB \neq BA$ in general, even if both exists.

Linear algebra: quick review

Some special matrices

- Square matrix: A is of size $n \times n$
- Diagonal matrix: $a_{ij} = 0$ if $i \neq j$.
- Identity matrix of order n: $I = [\delta_{ij}]$ where $\delta_{ij} = 1$ if $i = j$ and $= 0$ otherwise.
- Upper triangle matrix: $a_{ij} = 0$ if $i > j$.
- Lower triangle matrix: $a_{ij} = 0$ if $i < j$.

Linear algebra: quick review

Definition (Inverse of matrix)

An $n \times n$ matrix A is said to be **nonsingular** *(or* **invertible***) if there exists an $n \times n$ matrix, denoted by A^{-1}, such that $A(A^{-1}) = (A^{-1})A = I$. Here A^{-1} is called the* **inverse** *of matrix A.*

Definition

An $n \times n$ matrix A without an inverse is called **singular** *(or* **noninvertible***)*

Linear algebra: quick review

Several properties of inverse matrix:

- A^{-1} is unique.
- $(A^{-1})^{-1} = A$.
- If B is also nonsingular, then $(AB)^{-1} = B^{-1}A^{-1}$.

Linear algebra: quick review

Definition (Transpose)

The **transpose** *of an* $m \times n$ *matrix* $A = [a_{ij}]$ *is the* $n \times m$ *matrix* $A^\top = [a_{ji}]$.

Sometimes $A^\top$ is also written as $A^t, A', A^\top$.

- $(A^\top)^\top = A$
- $(AB)^\top = B^\top A^\top$
- $(A+B)^\top = A^\top + B^\top$
- If A is nonsingular, then $(A^{-1})^\top = (A^\top)^{-1}$.

Linear algebra: quick review

Definition (Determinant)

- *If* $A = [a]$ *is a* 1×1 *matrix, then* $\det(A) = a$.
- *If* A *is* $n \times n$ *where* $n > 1$, *then the* **minor** M_{ij} *is the determinant of the* $(n-1) \times (n-1)$ *submatrix of* A *by deleting its* i*th row and* j*th column.*
- *The* **cofactor** A_{ij} *associated with the minor* M_{ij} *is defined by* $A_{ij} = (-1)^{i+j} M_{ij}$.
- *The* **determinant** *of the* $n \times n$ *matrix* A, *denoted by* $\det(A)$ *(or* $|A|$*), is given by either of the followings:*

$$\det(A) = \sum_{j=1}^{n} a_{ij} A_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}, \quad \textit{for any } i = 1, \dots, n$$

$$\det(A) = \sum_{i=1}^{n} a_{ij} A_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}, \quad \textit{for any } j = 1, \dots, n$$

Linear algebra: quick review

Some properties of determinant

- If A has any zero row or column, then $\det(A) = 0$.
- If two rows (or columns) of A are the same, or one is a multiple of the other, then $\det(A) = 0$.
- Switching two rows (or columns) of A results in a matrix with determinant $-\det(A)$.
- Multiplying a row (or column) of A by λ results in a matrix with determinant $\lambda \det(A)$.
- $(E_i + \lambda E_j) \to E_i$ results in a matrix of the same determinant.

Linear algebra: quick review

Some properties of determinant

- $\det(AB) = \det(A)\det(B)$ if A and B are square matrices of same size.
- $\det(A^\top) = \det(A)$
- A is singular if any only if $\det(A) = 0$.
- If A is nonsingular, then $\det(A) \neq 0$ and $\det(A^{-1}) = \det(A)^{-1}$.
- If A is an upper or lower triangular matrix, then $\det(A) = \prod_{i=1}^{n} a_{ii}$.

Linear algebra: quick review

The following statements are equivalent:

- $Ax = 0$ has unique solution $x = 0$.
- $Ax = b$ has a unique solution for every b.
- A is nonsingular, i.e., A^{-1} exists.
- $\det(A) \neq 0$.

Matrix factorization

Gauss elimination can be used to compute **LU factorization** of a square matrix A:

$$A = LU$$

where L is a lower triangular matrix, and U is an upper triangular matrix.

Matrix factorization

If we have **LU factorization** of A, then

$$Ax = LUx = L(Ux) = b$$

so we solve x easily:

1. Solve y from $Ly = b$ by forward substitution;
2. Solve x from $Ux = y$ by backward substitution.

Total cost is $O(2n^2)$.

Matrix factorization

The cost reduction from $O(n^3/3)$ to $O(2n^2)$ is huge, especially for large n:

n	$n^3/3$	$2n^2$	% Reduction
10	$3.\bar{3} \times 10^2$	2×10^2	40
100	$3.\bar{3} \times 10^5$	2×10^4	94
1000	$3.\bar{3} \times 10^8$	2×10^6	99.4

Unfortunately, LU factorization itself requires $O(n^3)$ in general.

LU factorization

Now let's see how to obtain LU factorization by Gauss elimination.

Suppose we can perform Gauss elimination without any row exchange. In first round, we use a_{11} as the pivot and cancel each of $a_{21}, \dots, a_{n1}$ by

$$(E_j - m_{j1}E_1) \to E_j \quad \text{where } m_{j1} = \frac{a_{j1}}{a_{11}}, \quad j = 2, \dots, 4$$

This is equivalent to multiplying $M^{(1)}$ to A and get $A^{(2)} := M^{(1)}A$ where

$$M^{(1)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -m_{n1} & 0 & \cdots & 1 \end{bmatrix} \text{ and } A^{(2)} = \begin{bmatrix} a_{11} & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & * & \cdots & * \end{bmatrix}$$

LU factorization

In second round, we use current a_{22} as the pivot and cancel each of $a_{32}, \dots, a_{n2}$ by

$$(E_j - m_{j2}E_2) \to E_j \quad \text{where } m_{j2} = \frac{a_{j2}}{a_{22}}, \quad j = 3, \dots, 4$$

This is equivalent to multiplying $M^{(2)}$ to $A^{(2)}$ and get $A^{(3)} := M^{(2)}A^{(2)}$ where

$$M^{(2)} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -m_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & -m_{n2} & 0 & \cdots & 1 \end{bmatrix} \text{ and } A^{(3)} = \begin{bmatrix} a_{11} & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & * & \cdots & * \end{bmatrix}$$

LU factorization

When Gauss elimination finishes (total $n-1$ rounds), we will get an upper triangular matrix U:

$$U := M^{(n-1)}M^{(n-2)}\cdots M^{(1)}A$$

Define matrix L

$$L = (M^{(n-1)}M^{(n-2)}\cdots M^{(1)})^{-1} = (M^{(1)})^{-1}\cdots(M^{(n-2)})^{-1}(M^{(n-1)})^{-1}$$

Note that L is lower triangular (because each M is lower triangular, and inverse and product of lower triangular matrices are still lower triangular). So we get the LU factorization of A:

$$LU = (M^{(1)})^{-1}\cdots(M^{(n-2)})^{-1}(M^{(n-1)})^{-1}M^{(n-1)}M^{(n-2)}\cdots M^{(1)}A = A$$

LU factorization

It is easy to check that:

$$M^{(1)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -m_{n1} & 0 & \cdots & 1 \end{bmatrix} \text{ and } (M^{(1)})^{-1} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ m_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & 0 & \cdots & 1 \end{bmatrix}$$

$$M^{(2)} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -m_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & -m_{n2} & 0 & \cdots & 1 \end{bmatrix} \text{ and } (M^{(2)})^{-1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & m_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & m_{n2} & 0 & \cdots & 1 \end{bmatrix}$$

LU factorization

and finally there is

$$L = (M^{(1)})^{-1} \cdots (M^{(n-2)})^{-1}(M^{(n-1)})^{-1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ m_{21} & 1 & 0 & \cdots & 0 \\ m_{31} & m_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & m_{n3} & \cdots & 1 \end{bmatrix}$$

To summarize, the LU factorization of A gives L as above, and U as the result of Gauss elimination of A.

Gauss elimination row exchange

If Gauss elimination is done with row exchanges, then we will get LU factorization of PA where P is some row permutation matrix.

For example, to switch rows 2 and 4 of a 4×4 matrix A, the permutation matrix P is

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Some properties of permutation matrices:

- If P_1, P_2 are permutations, then $P_2 P_1$ is still permutation.
- $P^{-1} = P^\top$.

Diagonally dominate matrices

Now we consider two types of matrices for which Gauss elimination can be used effectively without row interchanges.

Definition (Diagonally dominate matrices)

An $n \times n$ matrix A is called **diagonally dominate** *if*

$$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|, \quad \forall i = 1, 2, \dots, n$$

An $n \times n$ matrix A is called **strictly diagonally dominate** *if*

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, \quad \forall i = 1, 2, \dots, n$$

Diagonally dominate matrices

Example

Consider the following matrices:

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 7 & 2 & 0 \\ 3 & 5 & -1 \\ 0 & 5 & -6 \end{bmatrix} \quad C = \begin{bmatrix} 6 & 4 & -3 \\ 4 & -2 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

A (and $A^\top$) is diagonally dominate, B is strictly diagonally dominate, $B^\top, C, C^\top$ are not diagonally dominate.

Diagonally dominate matrices

Theorem

If A is strictly diagonally dominant, then A is nonsingular. Moreover, Gauss elimination can be performed without row interchange to obtain the unique solution of $Ax = b$.

Diagonally dominate matrices

Proof.

If A is singular, then $Ax = 0$ has nonzero solution x. Suppose x_k is the component of x with largest magnitude:

$$|x_k| > 0 \quad \text{and} \quad |x_k| \geq |x_j|, \ \forall j \neq k$$

Then the product of x and the k-th row of A gives

$$a_{kk}x_k + \sum_{j \neq k} a_{kj}x_j = 0$$

From this we obtain

$$|a_{kk}| = \left| -\sum_{j \neq k} \frac{a_{kj}x_j}{x_k} \right| \leq \sum_{j \neq k} \frac{|x_j|}{|x_k|}|a_{kj}| \leq \sum_{j \neq k} |a_{kj}|$$

Contradiction. So A is nonsingular. $\square$

Diagonally dominate matrices

Proof (cont.) Now let's see how Gauss elimination works when A is strictly diagonally dominant. Consider 1st and ith ($i \geq 2$) rows of A:

$$|a_{11}| > \sum_{j \neq 1} |a_{1j}|, \quad |a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

If we perform $E_i - \frac{a_{i1}}{a_{11}} E_1 \to E_i$, the new values in row i are $a_{i1}^{(2)} = 0$ and $a_{ij}^{(2)} = a_{ij} - \frac{a_{i1}}{a_{11}} a_{1j}$ for $j \geq 2$. Therefore

$$\begin{aligned}
\sum_{\substack{j=2 \\ j \neq i}}^{n} |a_{ij}^{(2)}| &\leq \sum_{\substack{j=2 \\ j \neq i}}^{n} |a_{ij}| + \sum_{\substack{j=2 \\ j \neq i}}^{n} \left| \frac{a_{1j}}{a_{11}} \right| |a_{i1}| < |a_{ii}| - |a_{i1}| + \frac{|a_{11}| - |a_{1i}|}{|a_{11}|} |a_{i1}| \\
&= |a_{ii}| - \frac{|a_{1i}|}{|a_{11}|} |a_{i1}| \leq \left| |a_{ii}| - \frac{|a_{1i}|}{|a_{11}|} |a_{i1}| \right| = |a_{ii}^{(2)}|
\end{aligned}$$

As i is arbitrary, we know A remains strictly diagonally dominant after first round. By induction we know A stays as strictly diagonally dominant and Gauss elimination can be performed without row interexchange.

Positive definite matrices

Definition (Positive definite matrix)

A matrix A is called **positive definite** *(PD) if it is symmetric and $x^\top Ax > 0$ for any $x \neq 0$*

Remark

In some texts, A is called positive definite as long as $x^\top Ax > 0$ for any $x \neq 0$, so A is not necessarily symmetric. In these texts, the matrix in our definition above is called **symmetric positive definite** *(SPD).*

Positive definite matrices

We first have the following formula: if $x = (x_1, \dots, x_n)^\top$ and $A = [a_{ij}]$, then

$$x^\top A x = \sum_{i,j} a_{ij} x_i x_j$$

Positive definite matrices

Example

Show that the matrix A below is PD:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Solution: First A is symmetric. For any $x \in \mathbb{R}^3$, we have

$$\begin{aligned} x^\top A x &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 \\ &= x_1^2 + (x_1^2 - 2x_1x_2 + x_2^2) + (x_2^2 - 2x_2x_3 + x_3^2) + x_3^2 \\ &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2 \end{aligned}$$

Therefore $x^\top A x = 0$ if and only if $x_1 = x_2 = x_3 = 0$. So A is PD.

Positive definite matrices

Theorem

If A is an $n \times n$ positive definite matrix, then

- *A is nonsingular;*
- *$a_{ii} > 0$ for all i;*
- $\max_{i \neq j} |a_{ij}| \leq \max_i |a_{ii}|$;
- *$(a_{ij})^2 < a_{ii} a_{jj}$ for any $i \neq j$.*

Positive definite matrices

Proof.

- If $Ax = 0$, then $x^\top Ax = 0$ and hence $x = 0$ since A is PD. So A is nonsingular.
- Set $x = e_i$, where $e_i \in \mathbb{R}^n$ has 1 as the i-th component and zeros elsewhere. Then $x^\top Ax = e_i^\top Ae_i = a_{ii} > 0$.
- For any k, j, define $x, z \in \mathbb{R}^n$ such that $x_j = z_k = z_j = 1$ and $x_k = -1$, and $x_i = z_i = 0$ if $i \neq k, j$. Then we can show

  $$0 < x^\top Ax = a_{jj} + a_{kk} - a_{kj} - a_{jk}$$
  $$0 < z^\top Az = a_{jj} + a_{kk} + a_{kj} + a_{jk}$$

  Note that $a_{kj} = a_{jk}$, so we get $|a_{kj}| < \frac{a_{jj}+a_{kk}}{2} \leq \max_i a_{ii}$.
- For any $i \neq j$, set $x \in \mathbb{R}^n$ such that $x_i = \alpha$ and $x_j = 1$, and 0 elsewhere. Therefore $0 < x^\top Ax = a_{ii}\alpha^2 + 2a_{ij}\alpha + a_{jj}^2$ for any α. This implies that $4a_{ij}^2 - 4a_{ii}a_{jj} < 0$.

Positive definite matrices

Definition (Leading principal submatrix)

A **leading principal submatrix** *of A is the $k \times k$ upper left submatrix*

$$A_k = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{bmatrix}$$

Positive definite matrices

Theorem

A symmetric matrix A is PD if and only if every leading principal submatrix has a positive determinant.

Example

Use the Theorem above to check A is PD:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Positive definite matrices

Theorem

A matrix A is PD if and only either of the followings is true:

- *There exist a lower triangular matrix L with all 1 on its diagonal and a diagonal matrix D with all diagonal entries positive, such that $A = LDL^\top$.*
- *There exists a lower triangular matrix L with all diagonal entries positive such that $A = LL^\top$ (Cholesky factorization).*
- *Gauss elimination of A without row interchanges can be performed and all pivot elements are positive.*

Band matrices

Definition (Band matrix)

An $n \times n$ matrix A is called **band matrix** *if there exist p, q such that a_{ij} can be nonzero only if $i - q \leq j \leq i + p$. The band width is defined by $w = p + q + 1$.*

Definition (Tridiagonal matrix)

A band matrix with $p = q = 1$ is called **tridiagonal** *matrix.*

Crout factorization

The **Crout factorization** of a tridiagonal matrix is $A = LU$ where L is lower triangle, U is upper triangle, and both L, U are tridiagonal:

$$L = \begin{bmatrix} l_{11} & 0 & \cdots & 0 & 0 \\ l_{21} & l_{22} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & l_{n-1,n-1} & 0 \\ 0 & 0 & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix} \qquad U = \begin{bmatrix} 1 & u_{12} & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & u_{n-1,n} \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Note that a tridiagonal matrix A has $3n - 2$ unknowns, and the L and U together also have $3n - 2$ unknowns.

Crout factorization

Theorem

A tridiagonal matrix A has a Crout factorization if either of the following statements is true:

- *A is positive definite;*
- *A is strictly diagonally dominant;*
- *A is diagonally dominant,* $|a_{11}| > |a_{12}|$, $|a_{nn}| > |a_{n,n-1}|$, *and* $a_{i,i-1}, a_{i,i+1} \neq 0$ *for all* $i = 2, \ldots, n-1$.

Crout factorization

With the special form of A, L and U, we can obtain the Crout factorization $A = LU$ by solving l_{ij} ($i = 1, \dots, n$ and $j = i-1, i$) and $u_{i,i+1}$ ($i = 1, \dots, n-1$) from

$$\begin{aligned}
a_{11} &= l_{11} \\
a_{i,i-1} &= l_{i,i-1}, && \text{for } i = 2, \dots, n \\
a_{i,i} &= l_{i,i-1} u_{i-1,i} + l_{ii}, && \text{for } i = 2, \dots, n \\
a_{i,i+1} &= l_{ii} u_{i,i+1}, && \text{for } i = 1, \dots, n-1
\end{aligned}$$

When we use Crout factorization to solve $Ax = b$, the cost is only $5n - 4$ multiplications/divisions and $3n - 3$ additions/subtractions.

Vector norm

Definition

A **vector norm** *on* $\mathbb{R}^n$*, denoted by* $\|\cdot\|$*, is a mapping from* $\mathbb{R}^n$ *to* $\mathbb{R}$ *such that*

- $\|x\| \geq 0$ *for all* $x \in \mathbb{R}^n$*,*
- $\|x\| = 0$ *if and only if* $x = 0$*,*
- $\|\alpha x\| = |\alpha|\|x\|$ *for all* $\alpha \in \mathbb{R}$ *and* $x \in \mathbb{R}^n$*,*
- $\|x + y\| \leq \|x\| + \|y\|$ *for all* $x, y \in \mathbb{R}$*.*

Vector norm

Definition (l_p norms)

The l_p (sometimes L_p or ℓ_p) norm of a vector is defined by

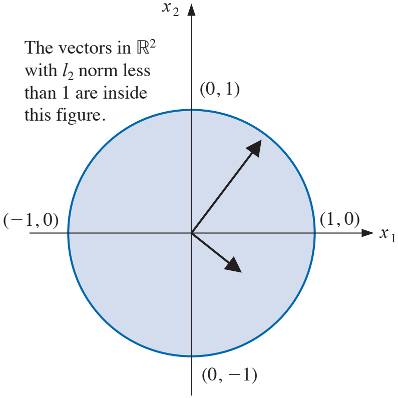
$$1 \le p < \infty : \qquad \|x\|_p = \Big(\sum_{i=1}^{n} |x_i|^p\Big)^{1/p}$$

$$p = \infty : \qquad \|x\|_\infty = \max_{1\le i\le n} |x_i|$$

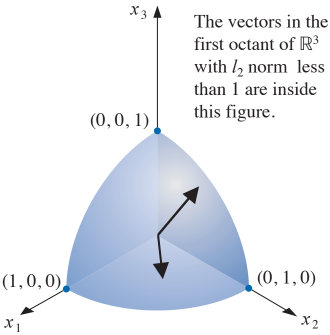
In particular, the l_2 norm is also called the **Euclidean norm***.*

Note that when $0 \le p < 1$, $\|\cdot\|_p$ is not norm, strictly speaking, but have some usages in specific applications.

l_2 norm

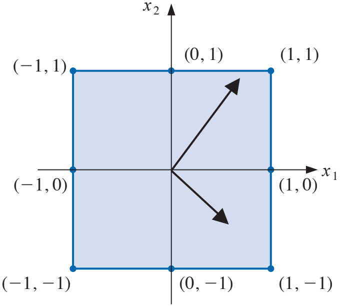


x_2
The vectors in $\mathbb{R}^2$ with l_2 norm less than 1 are inside this figure.
(0, 1)
(−1, 0)
(1, 0)
x_1
(0, −1)

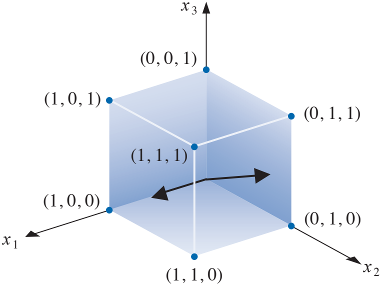


x_3
The vectors in the first octant of $\mathbb{R}^3$ with l_2 norm less than 1 are inside this figure.
(0, 0, 1)
(1, 0, 0)
(0, 1, 0)
x_1
x_2

l_∞ norm



The vectors in $\mathbb{R}^2$ with l_∞ norm less than 1 are inside this figure.



The vectors in the first octant of $\mathbb{R}^3$ with l_∞ norm less than 1 are inside this figure.

Vector norms

Example

Compute the l_2 and l_∞ norms of vector $x = (1, -1, 2) \in \mathbb{R}^3$.

Solution:

$$\|x\|_2 = \sqrt{|1|^2 + |-1|^2 + |2|^2} = \sqrt{6}$$

$$\|x\|_\infty = \max_{1 \le i \le 3} |x_i| = \max\{|1|, |-1|, |2|\} = 2$$

Theorem (Cauchy-Schwarz inequality)

For any vectors $x = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$ *and* $y = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$*, there is*

$$|x^\top y| = \Big|\sum_{i=1}^n x_i y_i\Big| \le \Big(\sum_{i=1}^n |x_i|^2\Big)^{1/2} \Big(\sum_{i=1}^n |y_i|^2\Big)^{1/2} = \|x\|_2 \|y\|_2$$

Proof.

It is obviously true for $x = 0$ or $y = 0$. If $x, y \neq 0$, then for any $\lambda \in \mathbb{R}$, there is

$$0 \le \|x - \lambda y\|_2^2 = \|x\|_2^2 - 2\lambda x^\top y + \lambda^2 \|y\|_2^2$$

and the equality holds when $\lambda = \|x\|_2 / \|y\|_2$. $\square$

Distance between vectors

Definition (Distance between two vectors)

The l_p **distance** *(*$1 \le p \le \infty$*) between two vectors* $x, y \in \mathbb{R}^n$ *is defined by* $\|x - y\|_p$*.*

Definition (Convergence of a sequence of vectors)

A sequence $\{x^{(k)}\}$ *is said to* **converge with respect to the** l_p **norm** *if for any given* $\epsilon > 0$*, there exists an integer* $N(\epsilon)$ *such that*

$$\|x^{(k)} - x\| < \epsilon, \quad \textit{for all } k \ge N(\epsilon)$$

Convergence of a sequence of vectors

Theorem

A sequence of vectors $\{x^{(k)}\}$ converges to x if and only if $x_i^{(k)} \to x_i$ for every $i = 1, 2, \dots, n$.

Theorem

For any vector $x \in \mathbb{R}^n$, there is

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty$$

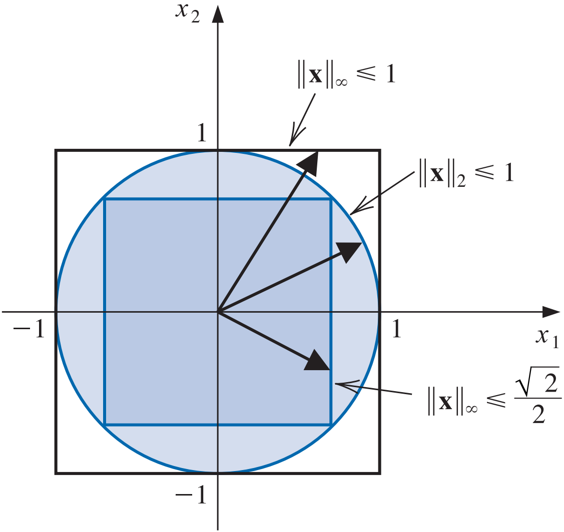
Proof.

$$\|x\|_\infty = \max_i |x_i| = \sqrt{\max_i |x_i|^2} \le \sqrt{|x_1|^2 + \cdots + |x_n|^2} = \|x\|_2$$

$$\begin{aligned}\|x\|_2 &= \sqrt{|x_1|^2 + \cdots + |x_n|^2} \le \sqrt{n \max_i |x_i|^2} \\ &= \sqrt{n}\sqrt{\max_i |x_i|^2} = \sqrt{n} \max_i |x_i| = \sqrt{n}\|x\|_\infty\end{aligned}$$

□

Compare l_2 and l_∞ norms in $\mathbb{R}^2$



Matrix norm

Definition

A **matrix norm** *on the set of* $n \times n$ *matrices is a real-valued function, denoted by* $\|\cdot\|$, *that satisfies the follows for all* $A, B \in \mathbb{R}^{n\times n}$ *and* $\alpha \in \mathbb{R}$*:*

- $\|A\| \geq 0$
- $\|A\| = 0$ *if and only if* $A = 0$ *the zero matrix,*
- $\|\alpha A\| = |\alpha|\|A\|$
- $\|A + B\| \leq \|A\| + \|B\|$
- $\|AB\| \leq \|A\|\|B\|$

Distance between matrices

Definition

Suppose $\|\cdot\|$ is a norm defined on $\mathbb{R}^{n\times n}$. Then the **distance between two** $n \times n$ **matrices** A **and** B *with respect to $\|\cdot\|$ is $\|A - B\|$ (check that it's a distance)*

Matrix norm can be induced by vector norms, and hence there are many choices. Here we focus on those induced by l_2 and l_∞ vector norms.

Matrix norm

Definition

If $\|\cdot\|$ *is a vector norm on* $\mathbb{R}^n$*, then the norm defined below*

$$\|A\| = \max_{\|x\|=1} \|Ax\|$$

is called the **matrix norm induced by vector norm** $\|\cdot\|$.

Matrix norm

Remark

- *Induced norms are also called natural norms of matrices.*
- *Unless otherwise specified, by matrix norms most books/papers refer to induced norms.*
- *The induced norm can be written equivalently as*

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

- *It can be easily extended to case $A \in \mathbb{R}^{m \times n}$.*

Matrix norm

Corollary

For any vector $x \in \mathbb{R}^n$, there is $\|Ax\| \leq \|A\| \|x\|$.

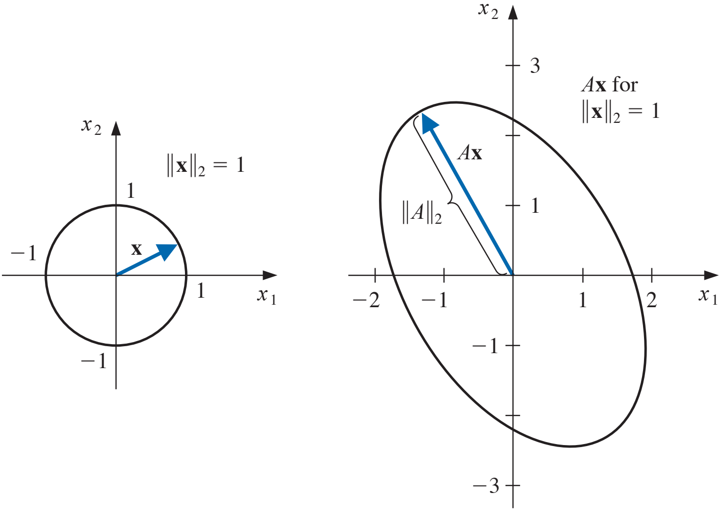
Proof.

It is obvious for $x = 0$. If $x \neq 0$, then

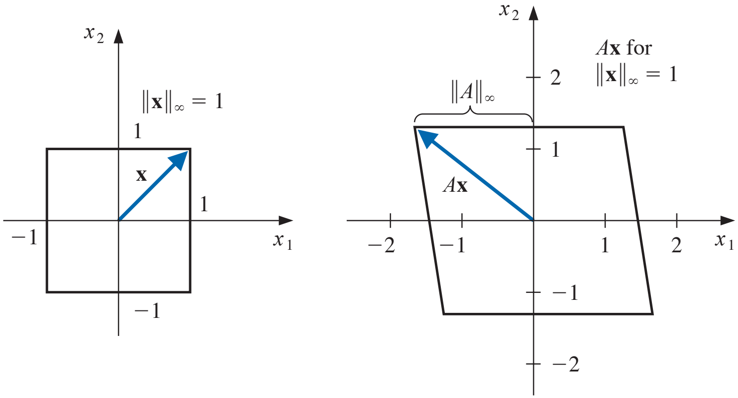
$$\frac{\|Ax\|}{\|x\|} \leq \max_{x' \neq 0} \frac{\|Ax'\|}{\|x'\|} = \|A\|$$

$\square$

Induced l_2 matrix norm



Induced l_∞ matrix norm



Matrix norm

Theorem

Suppose $A = [a_{ij}] \in \mathbb{R}^{n\times n}$*, then* $\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}|$.

Matrix norm

Proof.
For any x with $\|x\|_\infty = 1$, i.e., $\max_i |x_i| = 1$, there is

$$\begin{aligned}\|Ax\|_\infty &= \max\left\{\left|\sum_j a_{1j}x_j\right|, \dots, \left|\sum_j a_{nj}x_j\right|\right\} \\ &\le \max\left\{\sum_j |a_{1j}||x_j|, \dots, \sum_j |a_{nj}||x_j|\right\} \\ &\le \max\left\{\sum_j |a_{1j}|, \dots, \sum_j |a_{nj}|\right\} = \max_{1\le i\le n}\sum_{j=1}^n |a_{ij}|\end{aligned}$$

Suppose i' is such that $\sum_{j=1}^n |a_{i'j}| = \max_{1\le i\le n}\sum_{j=1}^n |a_{ij}|$, then by choosing $\hat{x}$ such that $\hat{x}_j = 1$ if $a_{i'j} \ge 0$ and -1 otherwise, we have $\sum_{j=1}^n a_{i'j}\hat{x}_j = \sum_{j=1}^n |a_{i'j}|$. So $\|A\hat{x}\|_\infty = \max_{1\le i\le n}\sum_{j=1}^n |a_{ij}|$. Note that $\|\hat{x}\|_\infty = 1$. Therefore $\|A\|_\infty = \max_{1\le i\le n}\sum_{j=1}^n |a_{ij}|$. □

Eigenvalues and eigenvectors of square matrices

Definition

The **characteristic polynomial** *of a square matrix* $A \in \mathbb{R}^{n \times n}$ *is defined by*

$$p(\lambda) = \det(A - \lambda I)$$

We call λ *an* **eigenvalue** *of* A *if* λ *is a root of* p, *i.e.,* $\det(A - \lambda I) = 0$. *Moreover, any nonzero solution* $x \in \mathbb{R}^n$ *of* $(A - \lambda I)x = 0$ *is called an* **eigenvector** *of* A *corresponding to the eigenvalue* λ.

Eigenvalues and eigenvectors of square matrices

Remark

- *$p(\lambda)$ is a polynomial of degree n, and hence has n roots.*
- *x is an eigenvector of A corresponding to eigenvalue λ iff $(A - \lambda I)x = 0$, i.e., $Ax = \lambda x$. This also means A applied to x is stretching x by λ.*
- *If x is an eigenvector of A corresponding to λ, so is αx for any $\alpha \neq 0$:*
$$A(\alpha x) = \alpha A x = \alpha \lambda x = \lambda(\alpha x)$$

Eigenvalues and eigenvectors of square matrices

Definition

Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $A \in \mathbb{R}^{n\times n}$, then the **spectral radius** *$\rho(A)$ is defined by $\rho(A) = \max_i |\lambda_i|$ where $|\cdot|$ is the absolute value (aka magnitude) of complex numbers.*

Eigenvalues and eigenvectors of square matrices

Some properties

Theorem

For a matrix $A \in \mathbb{R}^{n\times n}$, there are

- $\|A\|_2 = \sqrt{\rho(A^\top A)}$
- $\rho(A) \le \|A\|$ *for any norm* $\|\cdot\|$ *of* A

Proof.

- We later will show that both sides $= \sigma_1^2$, where σ_1 is the largest singular value of A.
- Let $\lambda := \rho(A)$ be the eigenvalue with largest magnitude. Then there exists eigenvector x such that

$$(\|A\| \ge) \frac{\|Ax\|}{\|x\|} = \frac{\|\lambda x\|}{\|x\|} = \frac{|\lambda|\|x\|}{\|x\|} = |\lambda|$$

□

Convergent matrix

Definition

A matrix $A \in \mathbb{R}^{n\times n}$ is said to be **convergent** *if*

$$\lim_{k\to\infty} A^k = 0$$

Theorem

The following statements are equivalent:

- *A is convergent.*
- $\lim_{k\to\infty} \|A^k\| = 0$ *for any norm* $\|\cdot\|$.
- $\rho(A) < 1$.
- $\lim_{k\to\infty} A^k x = 0$ *for any* $x \in \mathbb{R}$.

Jacobi iterative method

To solve x from $Ax = b$ where $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$, the **Jacobi iterative method** is

- Initialize $x^{(0)} \in \mathbb{R}^n$. Set $D = \mathrm{diag}(A)$, $R = A - D$.
- Repeat the following for $k = 0, 1, \dots$ until convergence:

$$x^{(k+1)} = D^{-1}(b - Rx^{(k)})$$

Remark

- *Needs nonzero diagonal entries, i.e., $a_{ii} \neq 0$ for all i.*
- *Usually faster convergence with larger $|a_{ii}|$.*
- *Stopping criterion can be $\frac{\|x^{(k)}-x^{(k-1)}\|}{\|x^{(k)}\|} \le \epsilon$ for some prescribed $\epsilon > 0$.*

Gauss-Seidel iterative method

To solve x from $Ax = b$ where $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$, the **Gauss-Seidel iterative method** is

- Initialize $x^{(0)} \in \mathbb{R}^n$. Set L to the lower triangular part (including diagonal) of A and $U = A - L$.
- Repeat the following for $k = 0, 1, \dots$ until convergence:

$$x^{(k+1)} = L^{-1}(b - Ux^{(k)})$$

Remark

- *Inverse of L requires forward substitution.*
- *Again needs nonzero diagonal entries, i.e., $a_{ii} \neq 0$ for all i.*
- *Stopping criterion can be $\frac{\|x^{(k)}-x^{(k-1)}\|}{\|x^{(k)}\|} \le \epsilon$ for some prescribed $\epsilon > 0$.*
- *Faster than Jacobi iterative method most of times.*

General iterative methods

Lemma
If $\rho(T) < 1$, then $(I - T)^{-1}$ exists and

$$(I - T)^{-1} = I - T + T^2 + \cdots = \sum_{j=0}^{\infty} T^j$$

General iterative methods

Proof.

We first show that $I - T$ is invertible, i.e., $(I - T)x = 0$ does not have nonzero solution. If not, then $\exists x \neq 0$ such that $(I - T)x = 0$, i.e., x is an eigenvector of $I - T$ corresponding to eigenvalue 0.

Notice that if $Tx = \lambda x$ if and only if $(I - T)x = (1 - \lambda)x$. So λ is an e.w. of T iff $1 - \lambda$ is an e.w. of $I - T$. Since $\rho(T) < 1$, we know $1 - \lambda \neq 0$, and hence $I - T$ cannot have 0 as e.w. Contradiction. So $I - T$ is invertible.

Define $S_m = I + T + T^2 + \cdots + T^m$. Then we can show $(I - T)S_m = I - T^{m+1}$. So due to $\rho(T) < 1$ we know

$$(I - T)\lim_{m\to\infty} S_m = \lim_{m\to\infty}(I - T)S_m = \lim_{m\to\infty}(I - T^{m+1}) = I$$

That is, $\sum_{m=0}^{\infty} T^m = \lim_{m\to\infty} S_m = (I - T)^{-1}$. $\square$

General iterative methods

A general iterative method has formula $x^{(k)} = Tx^{(k-1)} + c$ for $k = 1, 2, \dots$.

Example

- Jacobi iterative method:

$$x^{(k)} = D^{-1}(b - Rx^{(k-1)}) = -(D^{-1}R)x^{(k-1)} + D^{-1}b$$

  So $T = -D^{-1}R$ and $c = D^{-1}b$.

- Gauss-Seidel iterative method:

$$x^{(k)} = L^{-1}(b - Ux^{(k-1)}) = -(L^{-1}U)x^{(k-1)} + L^{-1}b$$

  So $T = -L^{-1}U$ and $c = L^{-1}b$.

General iterative methods

Theorem

For any initial $x^{(0)}$*, the sequence* $\{x^{(k)}\}_k$ *defined by* $x^{(k)} = Tx^{(k-1)} + c$ *converges to the unique solution of* $x = Tx + c$ *iff* $\rho(T) < 1$.

Proof.

$(\Leftarrow)$ Suppose $\rho(T) < 1$. Then

$$\begin{aligned} x^{(k)} &= Tx^{(k-1)} + c = T(Tx^{(k-2)} + c) + c \\ &= T^2x^{(k-2)} + (I + T)c \\ &= \cdots \\ &= T^kx^{(0)} + (I + T + T^2 + \cdots + T^k)c \end{aligned}$$

Let $k \to \infty$, we know $T^k \to 0$ and $(I + T + T^2 + \cdots + T^k) \to (I - T)^{-1}$, so $x^{(k)} \to (I - T)^{-1}c$, which is the unique solution of $x = Tx + c$. $\square$

General iterative methods

Theorem

For any initial $x^{(0)}$, *the sequence* $\{x^{(k)}\}_k$ *defined by* $x^{(k)} = Tx^{(k-1)} + c$ *converges to the unique solution of* $x = Tx + c$ *iff* $\rho(T) < 1$.

Proof.

($\Rightarrow$) Suppose for any initial $x^{(0)}$, the sequence $\{x^{(k)}\}_k$ defined by $x^{(k)} = Tx^{(k-1)} + c$ converges to the unique solution of $x = Tx + c$ iff $\rho(T) < 1$.
Let x^* be the solution of $x = Tx + c$. Then for any $z \in \mathbb{R}$, we set initial $x^{(0)} = x^* - z$. Then

$$\begin{aligned} x^* - x^{(k)} &= (Tx^* + c) - (Tx^{(k-1)} + c) = T(x^* - x^{(k-1)}) \\ &= \cdots = T^k(x^* - x^{(0)}) = T^k z \to 0 \end{aligned}$$

This implies $\rho(T) < 1$. $\square$

General iterative methods

Corollary

If $\|T\| < 1$ for any matrix norm $\|\cdot\|$, and c is given, then $\{x^{(k)}\}$ generated by $x^{(k)} = Tx^{(k-1)} + c$ converges to the unique solution x^* of $x = Tx + c$. Moreover

1. $\|x^* - x^{(k)}\| \leq \|T\|^k \|x^* - x^{(0)}\|$.
2. $\|x^* - x^{(k)}\| \leq \frac{\|T\|^k}{1-\|T\|} \|x^{(1)} - x^{(0)}\|$.

Proof.

1. Immediately following ($\Longrightarrow$) part of the theorem above.
2. Note that $\|x^* - x^{(1)}\| \leq \|T\| \|x^* - x^{(0)}\|$ and hence $\|x^{(1)} - x^{(0)}\| \geq \|x^* - x^{(0)}\| - \|x^* - x^{(1)}\| \geq (1 - \|T\|)\|x^* - x^{(0)}\|$.

$\square$

General iterative methods

Theorem

If A is strictly diagonally dominant, then from any initial $x^{(0)}$ both Jacobi and Gauss-Seidel iterative methods generate sequences that converge to the unique solution of $Ax = b$.

Relaxation techniques

The theory of general iterative methods suggest using a matrix T with smaller spectrum $\rho(T)$. To this end, we can use the relaxation technique to modify the iterative scheme.

- Original Gauss-Seidel iterative method:

$$x^{(k)} = -(L^{-1}U)x^{(k-1)} + L^{-1}b$$

- **Successive Over-Relaxation** (SOR) for Gauss-Seidel iterative method ($\omega > 1$):

$$x^{(k)} = (D - \omega L)^{-1}[(1-\omega)D + \omega U]x^{(k-1)} + \omega(D - \omega L)^{-1}b$$

where $D, -L, -U$ are the diagonal, strict lower, and strict upper triangular parts of A, respectively.

Relaxation techniques

Example

Compare Gauss-Seidel and SOR with $\omega = 1.25$, both using $x^{(0)} = (1, 1, 1)^\top$ as initial, to solve the system:

$$\begin{aligned} 4x_1 + 3x_2 &= 24 \\ 3x_1 + 4x_2 - x_3 &= 30 \\ -x_2 + 4x_3 &= -24 \end{aligned}$$

Relaxation techniques

Solution: Compare with true solution $(3, 4, -5)^\top$, we get:

Gauss-Seidel:

k	0	1	2	3	4	5	6	7
$x_1^{(k)}$	1	5.250000	3.1406250	3.0878906	3.0549316	3.0343323	3.0214577	3.0134110
$x_2^{(k)}$	1	3.812500	3.8828125	3.9267578	3.9542236	3.9713898	3.9821186	3.9888241
$x_3^{(k)}$	1	−5.046875	−5.0292969	−5.0183105	−5.0114441	−5.0071526	−5.0044703	−5.0027940

Successive Over-Relaxation:

k	0	1	2	3	4	5	6	7
$x_1^{(k)}$	1	6.312500	2.6223145	3.1333027	2.9570512	3.0037211	2.9963276	3.0000498
$x_2^{(k)}$	1	3.5195313	3.9585266	4.0102646	4.0074838	4.0029250	4.0009262	4.0002586
$x_3^{(k)}$	1	−6.6501465	−4.6004238	−5.0966863	−4.9734897	−5.0057135	−4.9982822	−5.0003486

The 5th iteration of SOR is better than 7th of GS.

Relaxation techniques

Theorem

If all diagonal entries of A are nonzero, then $\rho(T_\omega) \geq |\omega - 1|$, where $T_\omega = (D - \omega L)^{-1}[(1-\omega)D + \omega U]$.

This result says that SOR can converge only if $|\omega - 1| < 1$.

Relaxation techniques

Theorem

If A is positive definite and $|\omega - 1| < 1$, then the SOR converges starting from any initial $x^{(0)}$.

Theorem

If A is positive definite and tridiagonal, then $\rho(T_g) = [\rho(T_j)]^2 < 1$, where T_g and T_j are the T matrices of GS and Jacobi methods respectively, and the optimal ω for SOR is

$$\omega = \frac{2}{1 + \sqrt{1 - (\rho(T_j))^2}}$$

With this choice of ω, the spectrum $\rho(T_\omega) = \omega - 1$.

Iterative refinement

Definition (Residual)

Let $\tilde{x}$ be an approximation to the solution x of linear system $Ax = b$. Then $r = b - A\tilde{x}$ is called the **residual** *of approximation $\tilde{x}$.*

Remark

It seems intuitive that a small residual r implies a close approximation $\tilde{x}$ to x. However, it is not always true.

Iterative refinement

Example

The linear system $Ax = b$ is given by

$$\begin{bmatrix} 1 & 2 \\ 1.0001 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3.0001 \end{bmatrix}$$

has a unique solution $x = (1, 1)^\top$. Determine the residual vector r of a poor approximation $\tilde{x} = (3, -0.0001)^\top$.

Solution: The residual vector is

$$r = b - A\tilde{x} = \begin{bmatrix} 3 \\ 3.0001 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 1.0001 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -0.0001 \end{bmatrix} = \begin{bmatrix} 0.0002 \\ 0 \end{bmatrix}$$

So $\|r\|_\infty = 0.0002$ is small while $\|\tilde{x} - x\|_\infty = 2$ which still quite large.

Iterative refinement

Theorem

Suppose A is nonsingular, and $\tilde{x}$ is an approximation to the solution x of $Ax = b$, and $r = b - A\tilde{x}$ is the residual vector of $\tilde{x}$, then for any norm, there is

$$\|x - \tilde{x}\| \le \|r\| \cdot \|A^{-1}\|$$

Moreover, if $x \neq 0$ and $b \neq 0$, then there is

$$\frac{\|x - \tilde{x}\|}{\|x\|} \le \|A\| \cdot \|A^{-1}\| \cdot \frac{\|r\|}{\|b\|}$$

If $\|A\|\|A^{-1}\|$ is large, then small $\|r\|$ does not guarantee small $\|x - \tilde{x}\|$.

Iterative refinement

Proof.

Since x is a solution, we have $Ax = b$, we have $r = b - A\tilde{x} = Ax - A\tilde{x} = A(x - \tilde{x})$. Since A is nonsingular, we have $x - \tilde{x} = A^{-1}r$, and hence

$$\|x - \tilde{x}\| = \|A^{-1}r\| \le \|r\| \cdot \|A^{-1}\|$$

If $x \neq 0$ and $b \neq 0$, from $\|b\| = \|Ax\| \le \|A\| \cdot \|x\|$ we have $1/\|x\| \le \|A\|/\|b\|$. Multiplying this to the inequality above, we get

$$\frac{\|x - \tilde{x}\|}{\|x\|} \le \|A\| \cdot \|A^{-1}\| \cdot \frac{\|r\|}{\|b\|}$$

□

Iterative refinement

The number $\|A\| \cdot \|A^{-1}\|$ provide an indication between the error of approximation $\|x - \tilde{x}\|$ and size of residual r. So the larger $\|A\| \cdot \|A^{-1}\|$ is, the less power we have to control error using residual.

Definition (Condition number)

The **condition number** *of a nonsingular matrix A relative to a norm $\| \cdot \|_p$ is*

$$K_p(A) = \|A\|_p \cdot \|A^{-1}\|_p$$

The subscript p is often omitted if it's clear from context or it's not important.

Condition number

Remark

- *The condition number* $K(A) \geq 1$*:*

$$1 = \|I\| = \|AA^{-1}\| \leq \|A\| \cdot \|A^{-1}\| = K(A)$$

- *A matrix A is called* **well-conditioned** *if* $K(A)$ *is close to 1.*
- *A matrix A is called* **ill-conditioned** *if* $K(A) \gg 1$.

Condition number

Example

Determine the condition number of matrix

$$A = \begin{bmatrix} 1 & 2 \\ 1.0001 & 2 \end{bmatrix}$$

Condition number

Solution: Let's use l_∞ norm. Then

$$\|A\|_\infty = \max\{|1| + |2|, |1.0001| + |2|\} = 3.0001$$

Furthermore, there is

$$A^{-1} = \begin{bmatrix} -10000 & 10000 \\ 5000.5 & -5000 \end{bmatrix}$$

and hence $\|A^{-1}\|_\infty = 20000$. Therefore

$$K(A) = \|A\| \cdot \|A^{-1}\| = 3.0001 \times 20000 = 60002$$

Iterative refinement

Suppose $\tilde{x}$ is our current approximation to x. Let $\tilde{y} = x - \tilde{x}$, then $A\tilde{y} = A(x - \tilde{x}) = Ax - A\tilde{x} = b - A\tilde{x} = r$. If we can solve for $\tilde{y}$ here, we would get a new approximation $\tilde{x} + \tilde{y}$, expectedly to approximate x better.

This procedure is called **iterative refinement**.

Iterative refinement

Given A and b, Iterative Refinement first applies Gauss eliminations to $Ax = b$ and obtains approximation x.

Then, for each iteration $k = 1, 2, \dots, N$, do the following:

- Compute residual $r = b - Ax$;
- Solve y from $Ay = r$ using the same Gauss elimination steps.
- Set $x \leftarrow x + y$

The actual Iterative Refinement algorithm can also find approximation of condition number $K_\infty(A)$ (See textbook).

Perturbed linear system

In reality, A and b may be perturbed by noise or rounding errors δA and δb. Therefore, we are actually solving

$$(A + \delta A)x = b + \delta b$$

rather than $Ax = b$. This won't cause much issue if A is well-conditioned, but could be a problem otherwise.

Perturbed linear system

Theorem

Suppose A is nonsingular and $\|\delta A\| < \frac{1}{\|A^{-1}\|}$, then the solution $\tilde{x}$ of perturbed linear system $(A + \delta A)x = b + \delta b$ has an error estimate given by

$$\frac{\|x - \tilde{x}\|}{\|x\|} \le \frac{K(A)\|A\|}{\|A\| - K(A)\|\delta A\|}\left(\frac{\|\delta b\|}{\|b\|} + \frac{\|\delta A\|}{\|A\|}\right)$$

where x is the solution of the original linear system $Ax = b$.

Note that $K(A)\|\delta A\| = \|A\|\|A^{-1}\|\|\delta A\| < \|A\|$ so the denominator is positive.

Conjugate gradient method

Conjugate gradient (CG) method is particularly efficient for solving linear systems with large, sparse, and positive definite matrix A.

Equipped with proper preconditioning, CG can often reach very good result in $\sqrt{n}$ iterations (n the size of system).

The per-iteration cost is also low when A is sparse.

An alternate perspective of linear system

Theorem

Let A be positive definite, then x^* is the solution of $Ax = b$ iff x^* is the minimizer of

$$g(x) = \frac{1}{2}x^\top Ax - b^\top x$$

Proof.

Note that $\nabla g(x) = Ax - b$ and $\nabla^2 g(x) = A \succ 0$, so $g(x^*) = Ax^* - b = 0$ iff x^* is a minimizer of $g(x)$. $\square$

An alternate perspective of linear system

We have following observations:

- $r = b - Ax = -\nabla g(x)$ is the residual and also the steepest descent direction of $g(x)$ (recall that $\nabla g(x)$ is the steepest ascent direction).
- It seems intuitive to update $x \leftarrow x + t \cdot r = x - t\nabla g(x)$ with proper step size t.
- It turns out that we can find such t that makes the most progress.
- This method is called the "steepest descent method".
- However, it converges slowly and exhibits "zigzag" path for ill-conditioned A.

A-orthogonal

Conjugate gradient method amends this issue of steepest descent. To derive CG, we first present the following concept:

Definition

*Two vectors v and w are called A-**orthogonal** if $\langle v, Aw \rangle = 0$.*

Theorem

If A is positive definite, then there exists a set of independent vectors $\{v^{(1)}, \cdots, v^{(n)}\}$ such that $\langle v^{(i)}, Av^{(j)} \rangle = 0$ for all $i \neq j$.

Key idea of CG

Given previous estimate $x^{(k-1)}$ and a "search direction" $v^{(k)}$, CG will find scalars t_k and s_k to update x and v:

$$x^{(k)} = x^{(k-1)} + t_k v^{(k)}$$
$$v^{(k+1)} = r^{(k)} + s_k v^{(k)}$$

(where $r^{(k)} = b - Ax^{(k)}$), such that:

$$\langle v^{(k+1)}, Av^{(j)} \rangle = 0, \quad \forall j \le k$$
$$\langle r^{(k)}, v^{(j)} \rangle = 0, \quad \forall j \le k$$

If this can be done, then $\{v^{(1)}, \cdots, v^{(n)}\}$ is A-orthogonal.

Derivation of t_k and s_k

The main tool is mathematical induction: given $x^{(0)}$, first set $v^{(0)} = 0$, $r^{(0)} = b - Ax^{(0)}$, $v^{(1)} = r^{(0)}$. So

$$\langle v^{(k+1)}, Av^{(j)} \rangle = 0, \quad \forall j \le k$$
$$\langle r^{(k)}, v^{(j)} \rangle = 0, \quad \forall j \le k$$

is true for $k = 0$. Assume they hold for $k - 1$, we need to find t_k and s_k such that they also hold for k.

Derivation of t_k and s_k

We first find t_k: note that

$$r^{(k)} = b - Ax^{(k)} = b - A(x^{(k-1)} + t_k v^{(k)}) = r^{(k-1)} - t_k A v^{(k)}$$

Therefore, by induction hypothesis, there is

$$\begin{aligned} \langle r^{(k)}, v^{(j)} \rangle &= \langle r^{(k-1)} - t_k A v^{(k)}, v^{(j)} \rangle \\ &= \begin{cases} 0 & \text{if } j \le k-1, \\ \langle r^{(k-1)}, v^{(k)} \rangle - t_k \langle v^{(k)}, A v^{(k)} \rangle, & \text{if } j = k \end{cases} \end{aligned}$$

So we just need

$$t_k = \frac{\langle r^{(k-1)}, v^{(k)} \rangle}{\langle v^{(k)}, A v^{(k)} \rangle}$$

to make $\langle r^{(k)}, v^{(j)} \rangle = 0$.

Derivation of t_k and s_k

Then we find s_k: by the update of $v^{(k+1)}$, we have

$$\begin{aligned}\langle v^{(k+1)}, Av^{(j)}\rangle &= \langle r^{(k)} + s_k v^{(k)}, Av^{(j)}\rangle \\ &= \begin{cases} \langle r^{(k)}, Av^{(j)}\rangle, & \text{if } j \le k-1 \\ \langle r^{(k)}, Av^{(k)}\rangle + s_k\langle v^{(k)}, Av^{(k)}\rangle, & \text{if } j = k \end{cases}\end{aligned}$$

Note that $Av^{(j)} = \frac{Ax^{(j)} - Ax^{(j-1)}}{t_j} = \frac{r^{(j-1)} - r^{(j)}}{t_j}$, and $r^{(j-1)} - r^{(j)}$ is linear combination of $v^{(j-1)}, v^{(j)}, v^{(j+1)}$, so $\langle r^{(k)}, Av^{(j)}\rangle = 0$ for $j \le k-1$ due to induction hypothesis. Hence we just need

$$s_k = -\frac{\langle r^{(k)}, Av^{(k)}\rangle}{\langle v^{(k)}, Av^{(k)}\rangle}$$

to make $\langle r^{(k)}, Av^{(j)}\rangle = 0$ for all $j \le k$.

Derivation of t_k and s_k

We can further simply t_k and s_k:

Since that $v^{(k)} = r^{(k-1)} + s_{k-1} v^{(k-1)}$ and $\langle r^{(k-1)}, v^{(k-1)} \rangle = 0$, we have

$$t_k = \frac{\langle r^{(k-1)}, v^{(k)} \rangle}{\langle v^{(k)}, Av^{(k)} \rangle} = \frac{\langle r^{(k-1)}, r^{(k-1)} \rangle}{\langle v^{(k)}, Av^{(k)} \rangle}$$

Since $r^{(k-1)} = v^{(k)} - s_{k-1} v^{(k-1)}$, we have $\langle r^{(k)}, r^{(k-1)} \rangle = 0$. Since $Av^{(k)} = \frac{Ax^{(k)} - Ax^{(k-1)}}{t_k} = \frac{r^{(k-1)} - r^{(k)}}{t_k}$, we have $\langle r^{(k)}, Av^{(k)} \rangle = -\frac{\langle r^{(k)}, r^{(k)} \rangle}{t_k}$. Combining t_k expression above, we have

$$s_k = -\frac{\langle r^{(k)}, Av^{(k)} \rangle}{\langle v^{(k)}, Av^{(k)} \rangle} = -\frac{-\frac{\langle r^{(k)}, r^{(k)} \rangle}{t_k}}{\frac{\langle r^{(k-1)}, r^{(k-1)} \rangle}{t_k}} = \frac{\langle r^{(k)}, r^{(k)} \rangle}{\langle r^{(k-1)}, r^{(k-1)} \rangle}$$

Conjugate gradient method

Since $\langle r^{(n)}, v^{(k)} \rangle = 0$ for all $k = 1, \ldots, n$ and the A-orthogonal set $\{v^{(1)}, \cdots, v^{(n)}\}$ is independent when A is positive definite, we know $r^{(n)} = b - Ax^{(n)} = 0$, i.e., $x^{(n)}$ is the solution.

This shows that CG converges in at most n steps, assuming all arithmetics are exact.

Conjugate gradient method

- Input: $x^{(0)}$, $r^{(0)} = b - Ax^{(0)}$, $v^{(1)} = r^{(0)}$.
- Repeat the following for $k = 1, \dots, n$ until $r^{(k)} = 0$:

$$
\begin{aligned}
t_k &= \frac{\langle r^{(k-1)}, r^{(k-1)} \rangle}{\langle v^{(k)}, Av^{(k)} \rangle} \\
x^{(k)} &= x^{(k-1)} + t_k v^{(k)} \\
r^{(k)} &= r^{(k-1)} - t_k A v^{(k)} \\
s_k &= \frac{\langle r^{(k)}, r^{(k)} \rangle}{\langle r^{(k-1)}, r^{(k-1)} \rangle} \\
v^{(k+1)} &= r^{(k)} + s_k v^{(k)}
\end{aligned}
$$

- Output: $x^{(k)}$.

Preconditioning

The convergence rate of CG can be greatly improved by **preconditioning**. Preconditioning reduces condition number of A first if A is ill-conditioned. With preconditioning, CG usually converges in $\sqrt{n}$ steps.

The preconditioning is done by using some nonsingular matrix C, we can get $\tilde{A} = C^{-1}A(C^{-1})^\top$ such that $K(\tilde{A}) \ll K(A)$.

Now by defining $\tilde{x} = C^\top x$ and $\tilde{b} = C^{-1}b$, we obtain a new linear system $\tilde{A}\tilde{x} = \tilde{b}$, which is equivalent to $Ax = b$. Then we can apply CG to the new system $\tilde{A}\tilde{x} = \tilde{b}$.

Preconditioner

There are various methods to choose the preconditioner C.

- Choose $C = \text{diag}(\sqrt{a_{11}}, \dots, \sqrt{a_{nn}})$.
- Approximate Cholesky's factorization $LL^\top \approx A$ (by ignoring small values in A) and set $C = L$ (then $C^{-1}A(C^{-1})^\top \approx L^{-1}(LL^\top)L^{-T} = I$).
- Many others...

Preconditioned conjugate gradient method

- Input: Preconditioner C, $x^{(0)}$, $r^{(0)} = b - Ax^{(0)}$, $w^{(0)} = C^{-1}r^{(0)}$, $v^{(1)} = C^{-T}w^{(0)}$.
- Repeat the following for $k = 1, \dots, n$ until $r^{(k)} = 0$:

$$
\begin{aligned}
\tilde{t}_k &= \frac{\langle w^{(k-1)}, w^{(k-1)} \rangle}{\langle v^{(k)}, Av^{(k)} \rangle} \\
x^{(k)} &= x^{(k-1)} + \tilde{t}_k v^{(k)} \\
r^{(k)} &= r^{(k-1)} - \tilde{t}_k A v^{(k)} \\
w^{(k)} &= C^{-1} r^{(k)} \\
\tilde{s}_k &= \frac{\langle w^{(k)}, w^{(k)} \rangle}{\langle w^{(k-1)}, w^{(k-1)} \rangle} \\
v^{(k+1)} &= C^{-\top} w^{(k)} + \tilde{s}_k v^{(k)}
\end{aligned}
$$

- Output: $x^{(k)}$.

A comparison

Example

Given A and b below, we use the methods above to solve $Ax = b$.

$$A = \begin{bmatrix} 0.2 & 0.1 & 1 & 1 & 0 \\ 0.1 & 4 & -1 & 1 & -1 \\ 1 & -1 & 60 & 0 & -2 \\ 1 & 1 & 0 & 8 & 4 \\ 0 & -1 & -2 & 4 & 700 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

True solution is

$$x^* = \begin{bmatrix} 7.859713071 \\ 0.4229264082 \\ -0.07359223906 \\ -0.5406430164 \\ 0.01062616286 \end{bmatrix}$$

A comparison

A comparison of Jacobi, Gauss-Seidel, SOR, CG, and PCG on the problem above.

Method	Number of Iterations	$\mathbf{x}^{(k)}$	$\|\mathbf{x}^* - \mathbf{x}^{(k)}\|_\infty$
Jacobi	49	$(7.86277141, 0.42320802, -0.07348669, -0.53975964, 0.01062847)^t$	0.00305834
Gauss-Seidel	15	$(7.83525748, 0.42257868, -0.07319124, -0.53753055, 0.01060903)^t$	0.02445559
SOR ($\omega = 1.25$)	7	$(7.85152706, 0.42277371, -0.07348303, -0.53978369, 0.01062286)^t$	0.00818607
Conjugate Gradient	5	$(7.85341523, 0.42298677, -0.07347963, -0.53987920, 0.008628916)^t$	0.00629785
Conjugate Gradient (Preconditioned)	4	$(7.85968827, 0.42288329, -0.07359878, -0.54063200, 0.01064344)^t$	0.00009312